Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology Using Facial Expression Recognition and an NLP-Based Chatbot

INDIVIDUAL THESIS

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Dual-Engine 3D/2D Sign Avatar Lecture Generator and ASL Fingerspelling Studio

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October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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Supervisor certification. The candidate carried out this undergraduate research study toward the Bachelor of Science (Honors) degree in Information Technology under our direct supervision and guidance.

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ABSTRACT

Deaf and hard-of-hearing (DHH) students studying O/L Information and Communication Technology often face challenges while learning with computers. Mainstream Learning Management Systems use auditory lectures and unstructured text transcripts which creates a friction for many deaf learners who experience written natural language as a second language. This friction is exacerbated when learning technical STEM topics where new technical terms and computing acronyms do not have formal manual signs in national sign languages such as Sri Lankan Sign Language and American Sign Language. Mainstream e-learning software fails, omits or displays raw text when faced with such out-of-vocabulary technical terms resulting in split visual attention for such DHH learners.

This thesis presents the design, implementation and evaluation of SignLearn AI, an intelligent dual mode assistive learning platform developed as part of the R26-IT-18 Accessible Emotion-Aware E-Learning Platform. The system introduces two novel learning interfaces:

1. An ASL Fingerspelling Studio that uses spaCy natural language processing and curricular regular expression filters to identify technical ICT keywords without existing standard gestures (eg: RAM, CPU, ALU, SQL, BIOS, HTTP), animate a 3D articulated hand rig using Quaternions SLERP at a stable 60FPS, provide an interactive letter by letter timeline stepper, detailed anatomical handshape guidance cards and an in-browser canvas media stream capture pipeline that exports keyword demonstrations as 1080p 60FPS videos for offline revision.
2. A 2D Canvas Avatar Keyword Lecture Player driven by an 88 frame WLASL dataset pipeline with color coded dual-hand articulated skeletal rigs, variable speed playback (0.25x to 1.25x), gesture fallback states and synchronized bilingual English-Sinhala subtitles (eg: පරිගණකය for Computer).

The platform was evaluated using computational frame profiling, linguistic classification benchmarks and user evaluation with 18 DHH learners and 5 special education instructors. The automated keyword filter achieved a 96.4% macro F1-score, the 3D hand rig achieved a stable 60FPS on standard hardware and in-browser video compilation achieved full 1080p video clips in under 1.5s without server compute. The learners demonstrated a 43.8% increase in technical acronym retention and reported significantly reduced visual fatigue, demonstrating the value of SignLearn AI as a practical assistive technical education solution.

Keywords: Deaf Education, 3D Sign Avatar, ASL Fingerspelling Studio, Out-of-Vocabulary ICT Keywords, Canvas Stream Video Export, WLASL, Dual-Coding Theory, O/L ICT Accessibility.

ACKNOWLEDGEMENTS

I would like to express my sincere appreciation to my research supervisor Mr. Dhammika De Silva, and the co-supervisor Mr. Samantha Rajapaksha; their guidance, technical advise, and encouragement throughout the design and realization of my individual research deliverable, SignLearn AI: 3D Sign Avatar Lecture Generator and ASL Fingerspelling Studio, played a major role in enhancing the technical aspects of this work.

The feedback on computer graphics, web architectures, and accessibility design from my supervisors, contributed significantly towards strengthening the technical foundations of this work. Similarly, I extend my gratitude to the Sri Lanka Institute of Information Technology (SLIIT) and the Faculty of Computing for the research infrastructure, laboratory facilities, and academic guidance provided that contributed towards the realisation of this project. I would also like to express my gratitude to the administrative leadership, specialized instructors, and the deaf students from participating special education units in Sri Lanka for sharing their time, patience, and feedback during our field trials.

I am grateful to my research colleagues in group R26-IT-18; Tharaka W.A.K.N., Kothalawala L.S.S., and Nawarathna I.K.K.R.B. for collaborating closely in order to coordinate our microservice interfaces and communication pipelines. Lastly, I extend my deepest appreciation to my parents, family, and close friends for their patient support and encouragement throughout my undergraduate studies.

TABLE OF CONTENTS

1 INTRODUCTION 1
1.1 Project Context & R26-IT-18 Platform Ecosystem 1
1.2 Individual Research Focus: SignLearn AI 2
1.3 Problem Statement 3
1.4 Research Gap: The Lexical Void in Technical Sign Languages 3
1.5 Aim and Objectives 4
1.6 Research Questions and Hypotheses 4
1.7 Scope, Limitations, and Assumptions 5
1.8 Summary of Individual Contributions 5
1.9 Thesis Organization 6

2 LITERATURE REVIEW AND RELATED 7
2.1 Review Method and Scope 7
2.2 Cognitive Foundations: Dual-Coding and Split-Attention Effects 7
2.3 Sign Language Avatars: From Concatenated Video to Skeletal Rigs 7
2.4 Handling Out-of-Vocabulary Acronyms and Technical Fingerspelling 8
2.5 Real-Time Client-Side Media Streams and Browser Video Export 8
2.6 Comparative Analysis and Conceptual Framework 9

3 RESEARCH METHODOLOGY 10
3.1 Overall Architectural Design and Processing Flow 10
3.2 Automated NLP Keyword Extraction and Lexicon Verification Gate 10
3.3 Kinematic Modeling: 3D Articulated Hand Rig and 2D Skeletal Tracking 13
3.4 Interactive Studio Engineering: Timeline Stepper, Speed Controls, and Guides 14
3.5 WebGL Canvas Stream Recording and Client-Side Video Export 14
3.6 Evaluation Metrics and Experimental Setup 14
3.7 Research Ethics and Accessibility Considerations 15

4 SYSTEM ARCHITECTURE AND DETAILED DESIGN 16
4.1 Functional and Non-Functional Requirements 16
4.2 Platform Microservices Integration within R26-IT-18 17
4.3 Design Decisions and Architectural Trade-Offs 17
4.4 User Interface Architecture: Studio vs. Lecture Player 18
4.5 Data Contracts and API Endpoint Specifications 18

5 SYSTEM IMPLEMENTATION AND TESTING 19
5.1 Technology Stack and Tooling Specifications 19
5.2 Implementation of Core Rendering and Recording Engines 19
5.3 Microservice Coordination and WebSocket Telemetry 20
5.4 Verification Testing and Test Case Matrix 21
5.5 Containerization and System Deployment 21

6 RESULTS AND EMPIRICAL VALIDATION 22

6.1 Linguistic Keyword Extraction Accuracy 22
6.2 Rendering Performance, Frame Latency, and Stability 22
6.3 Canvas MediaRecorder Encoding Profiling 23
6.4 Usability and Pedagogical Validation with DHH Learners 24
6.5 Summary of Empirical Findings 25
7 DISCUSSION 26
7.1 Interpretation of Theoretical and Empirical Findings 26
7.2 Comparison with Literature Baselines and Existing Solutions 26
7.3 Research Contributions 26
7.4 Product and Architectural Contributions 26
7.5 Limitations and Threats to Validity 27
7.6 Pedagogical and Policy Implications 27
8 COMMERCIALISATION AND SUSTAINABILITY 28
8.1 Value Proposition and Target Market 28
8.2 Market Analysis and Competitive Positioning 28
8.3 Business Model, Pricing Tiers, and SaaS Scalability 28
8.4 Intellectual Property, Compliance, and Standards 29
8.5 Alignment with UN Sustainable Development Goals (SDGs) 29
9 CONCLUSION AND FUTURE WORK 30
9.1 Summary of the Study 30
9.2 Fulfillment of Research Objectives 30
9.3 Practical Impact 30
9.4 Future Work and Research Directions 31
REFERENCES 32
APPENDICES 34
Appendix A: Complete 26-Letter ASL Posture Angular Definition Table 34
Appendix B: Extended Browser and Device Execution Compatibility Log 36
Appendix C: Usability Evaluation Instrument and Teacher Assessment Protocol 37
Appendix D: Academic Integrity and AI Disclosure Statement 38

LIST OF FIGURES

Figure 3.1: Dual-Stream Technical Adaptation Pipeline: Text Ingestion, NLP Triage, Dual-Engine Animation, and Client Media Export 10

Figure 3.2: Dual-Dictionary Lexical Disambiguation Gate and Mode Dispatch Logic 11

Figure 3.3: Spherical Linear Interpolation (SLERP) Between Two Bone Rotations 13

Figure 4.1: Inter-Module Platform Telemetry and Central WebSocket Bus (R26-IT-18 Architecture) 17

Figure 6.1: Frame Rate Performance Across Benchmark Device Tiers 23

Figure 8.1: Strategic Value Proposition and Target Market Matrix 28

LIST OF TABLES

Table 2.1: Architectural Comparison Across Assistive Sign Technologies 9
Table 3.1: Technical O/L ICT Domain Terminology Classified by Linguistic Mode 12
Table 4.1: Component RESTful API Endpoints 18
Table 5.1: System Technology Stack 19
Table 5.2: Functional Test Matrix 21
Table 6.1: Performance of the Automated Keyword Identification Engine 22
Table 6.2: Frame Rate, Computation Time, and Memory Benchmarks Across Device Tiers 23
Table 6.3: Client-Side 1080p 60FPS Video Export Performance 24
Table 6.4: DHH Learner Usability and Comprehension Assessment Results 24
Table A.1: 26-Letter ASL Manual Alphabet Geometric Mapping and Anatomical Posture Descriptions 34
Table B.1: Extended Browser and Device Execution Compatibility Log 36

LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
ASL	American Sign Language
CLT	Cognitive Load Theory
CPU	Central Processing Unit
DHH	Deaf and Hard-of-Hearing
FK	Forward Kinematics
FPS	Frames Per Second
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
ICT	Information and Communication Technology
IK	Inverse Kinematics
JSON	JavaScript Object Notation
LMS	Learning Management System
NLP	Natural Language Processing
O/L	Ordinary Level
OOV	Out-Of-Vocabulary
RAM	Random Access Memory
SDG	Sustainable Development Goal
SDK	Software Development Kit
SLERP	Spherical Linear Interpolation
SLSL	Sri Lankan Sign Language
SUS	System Usability Scale
UI/UX	User Interface / User Experience
WCAG	Web Content Accessibility Guidelines
WebGL	Web Graphics Library
WLASL	Word-Level American Sign Language

1 INTRODUCTION

1.1 Project Context & R26-IT-18 Platform Ecosystem

Online and distance education systems have become integral aspects to modern secondary school curricula. However, existing learning management systems (LMS) and multimedia lecture repositories are predominantly optimized for hearing students. Their instructional architecture is centered around spoken narrative lectures, rapid conversational delivery, auditory cues and unstructured text closed captions. For students who are deaf or hard-of-hearing (DHH), this audio-centric approach poses additional educational barriers. Particularly acute in developing countries like Sri Lanka, secondary students preparing for National General Certificate of Education Ordinary Level (O/L) examinations do not have access to specialized digital learning environments.

The O/L Information and Communication Technology (ICT) syllabus requires that students understand technical and abstract concepts including computer system architecture, internal processing cycles, volatile and non-volatile storage hierarchies, database schema normalization, telecommunications models and procedural logic. When taught by standard video presentations and static electronic texts, DHH students are forced to continuously divide their visual attention between reading unsegmented technical subtitles and observing rapid graphical demonstrations on screen, violating principles of split-attention learning.

To address these systemic barriers, research group R26-IT-18 designed and developed the Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology. The platform unifies five modular, interoperable microservices in a coherant pedagogical architecture:

1. A Real-Time Attention Monitoring and Telemetry Module which utilizes commodity webcams to calculate Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), PERCLOS drowsiness metrics, and head-pose gaze coordinates to determine when a student is losing focus during a lesson.
2. A Knowledge Graph-Driven Question Generation System (GQSA) which utilizes interconnected topic graphs of the national O/L ICT curriculum to

provide targeted multiple-choice diagnostic popups when a student misses a concept.

3. An Emotion-Aware Adaptive Conversational Agent (EARA) which provides on-demand explanations, past-paper revision assistance and text simplifications specific to the learner's emotional and attentional state.
4. SignLearn AI: The Dual-Mode 3D/2D Sign Avatar Lecture Generator and ASL Fingerspelling Studio, which is able to translate technical instructional sentences into fluid sign representations.
5. Finally, a Smart ESP32 Haptic Wristband and Sign Practice Module which is able to provide discreet, non-auditory vibrational pulses to notify distracted students and allow for camera-based sign practice.

Together, these five modules form a closed-loop assistive educational framework; the e-learning platform observes students' focus, identifies concepts that they have missed, and provides non-auditory, visually grounded instructional feedback

1.2 Individual Research Focus: SignLearn AI

Within this platform, my individual research and engineering deliverable is SignLearn AI: The Dual-Mode 3D/2D Sign Avatar Lecture Generator and ASL Fingerspelling Studio (Student ID: IT22197764). While computer vision modules detect when attention lapses and knowledge graphs what concept requires remediation, SignLearn AI serves as the primary communication engine governing how technical lesson materials are delivered to the deaf learner.

Deaf people communicate primary through visual-spatial sign language. Natural written English is processed through grammatical translation rather than phonetic sound, making dense technical text slow and difficult to read. SignLearn AI solves this problem with two specialized, complementary tools:

1. **The ASL Fingerspelling Studio** was designed for technical computing terms/acronyms that do not have standardized gestures. It features an Articulated 3D Hand Rig in Three.js, 26-letter interactive alphabet matrix, anatomical posture guidance cards, an interactive timeline stepper, and an in-browser media recorder that exports demonstrations as downloadable 1080p 60FPS video files for offline study.

2. **The Keyword Lecture Player** is built for continuous lesson text, parsing instructional notes, extracting key terms, and matching to an 88-frame WLASL dataset using dual articulated 2D hand rigs with color-coded finger tracking, variable speed (0.25×-1.25×), and synchronized bilingual English-Sinhala subtitles.

1.3 Problem Statement

Deaf O/L ICT students encounter challenges while reading abstract computing concepts. Reading technical subtitles in fast-paced manner combined with watching subtitled visual diagrams creates a cognitive burden of split-attention visual fatigue. However, the existing sign language tools are limited by static dictionaries. If the lesson discusses new hardware components, network protocols or cybersecurity threats, computer signs cannot be used to sign material not in their dictionary.

This raises the main research question of this study:

How might an automated e-learning sub-system recognize out-of-vocabulary ICT keywords in English instructional sentences, synthesize smooth 3D skeletal hand animations in real-time, and give deaf students an interactive studio to control playback and export such demonstrations as downloadable revision videos?

1.4 Research Gap: The Lexical Void in Technical Sign Languages

A look into the current state of educational assistive technologies reveal a particular dearth of technical lexical items in sign language.

1. There are no standard gestures for the growing number of technical terms found in the computer science field of study.

   Computer related terms such as Cache Memory, Motherboard, Malware, Phishing, Bandwidth, RAM, and Defragmentation are not included in the sign language dictionaries like SSL or ASL.

2. Avatar's encounter difficulties when coming across out-of-lexicon technical terms These avatars either stumble over unknown linguistic items, omit them in the video, or replace them with similar looking gestures.
3. There are no automated keyword extraction algorithms that allow an instructor to specifically identify the target sign words for the video. In the past, the

creation of an educational video involved a human translator who identified words that would need to be translated into fingerspelling.

4. The application was also devoid of downloadable assets such as videos or images that could be used by the student as a guide or mnemonic devise for practicing the target signs after the video. Students studying these videos had to rely exclusively on the signs performed in the videos.

1.5 Aim and Objectives

Aim: To design, construct, and evaluate an intelligent dual-mode 3D/2D Sign Avatar Lecture Generator that detects out-of-vocabulary ICT keywords from teaching sentences, generates high-quality 3D fingerspelling animations, and allows students to export these demonstrators as downloadable 1080p video files.

Specific Objectives:

1. Develop a Natural Language Processing (NLP) keywords extraction engine that analyzes instructions, matches terms against signs dictionary, and flags technical ICT terms without manual signs.
2. Construct an articulated 3D character hand rig in Three.js and WebGL that renders fingerspelling animations at 60 frames per second using Quaternion Spherical Linear Interpolation (SLERP).
3. Develop an interactive Fingerspelling Studio with variable playback speed options (0.25x to 1.25x), camera control orbit, and loop segments.
4. Implement an in-browser media capture pipeline that can record avatar animations and export them as downloadable 1080p MP4/WebM video files.
5. Construct a dual-hand articulated 2D canvas lecture video player that supports WLASL gestures, back-up signs, and bilingual English-Sinhala subtitles.
6. Integrate the module into the central R26-IT-18 platform architecture.
7. Benchmark computation performance and conduct user evaluations with DHH students and educators to assess pedagogical effectiveness.

1.6 Research Questions and Hypotheses

RQ1: Can an NLP rule-and-lexicon pipeline reliably detect technical ICT keywords without corresponding native sign translations from English sentences with high precision?

RQ2: Does quaternion-based bone interpolation produce natural, comprehensible 3D hand fingerspelling animations on consumer web browsers without frame drops?

RQ3: Does the ability to interact with a 3D Fingerspelling Studio and download individual video clips improve technical vocabulary retention among deaf learners?

Hypothesis: Automated identification and 3D fingerspelling animation of out-of-vocabulary ICT keywords, coupled with interactive playback controls and video export capabilities, significantly enhances technical vocabulary retention and learning autonomy relative to static text transcripts.

1.7 Scope, Limitations, and Assumptions

Scope: The system is focused on the Sri Lankan G.C.E. O/L Information and Communication Technology curriculum (Grades 10–11). The fingerspelling studio implements the 26 Latin alphabet letters and the Arabic numerals (0–9) with a base dictionary of 30 essential ICT terms and presets spanning 5 curriculum modules (CPU & Memory, Internet & Web, Relational Databases, Hardware & Boot, Cybersecurity).

Limitations: The video rendering pipeline executes client-side and is dependent on the device's GPU performance for WebGL canvas recording.

Assumptions: Users have access to modern Chromium-based web browsers that support WebGL 2.0 and the MediaStream Recording API.

1.8 Summary of Individual Contributions

- My specific contributions to this research include the following:
- Formulation of the lexical void problem for technical ICT education in deaf e-learning platforms
- Development of the NLP keyword identification engine that classifies words into standard signs or fingerspelled terms
- Design of the 3D hand skeletal rig and quaternion interpolation logic in Three.js
- Engineering of the interactive Fingerspelling Studio interface with speed regulation, camera repositioning, timeline stepping, and anatomical posture guides

- Development of the 2D Canvas Keyword Lecture Player with two articulated hand rigs, color-coded finger tracking, and bilingual EnglishSinhala subtitles
- Implementation of the client-side canvas recording pipeline that produces downloadable 1080p 60FPS video files
- Validation of the system through performance profiling and usability evaluations with deaf students and educators

1.9 Thesis Organization

Following the official SLIIT thesis format below, my dissertation will cover the following chapters:

Chapter 2 will provide a literature review regarding sign language avatar, fingerspelling mechanisms, and cognitive theories.

Chapter 3 is devoted to explaining the methodology, keyword recognition algorithms, quaternion mathematics, and video export procedures.

The system architecture, component interface, and UI design are presented in Chapter 4.

In Chapter 5, the implementation details, Docker, and experimental results are provided.

The experimental results, export procedures, and video analysis are given in Chapter 6.

Theoretical and practical implications of the research results are given in Chapter 7.

Commercial aspects, licensing models, and SDG principles are discussed in Chapter 8.

The final chapter concludes the research and suggests some future improvements.

2 LITERATURE REVIEW AND RELATED

2.1 Review Method and Scope

This literature review considers peer-reviewed publications found in IEEE Xplore, ACM Digital Library, SpringerLink, and ScienceDirect databases. The years of publication are between 2018 and 2026. The research areas include educational psychology, 3D computer graphics, Natural Language Processing, sign language avatar, manual fingerspelling, out-of-vocabulary, and browser-based graphics pipeline.

2.2 Cognitive Foundations: Dual-Coding and Split-Attention Effects

The pedagogical foundation of this research is Allan Paivio's Dual-Coding Theory (DCT) and John Sweller's Cognitive Load Theory (CLT). DCT indicates that working memory processes information through two separate channels: a verbal/symbolic channel (handling text and speech) and a non-verbal/pictorial channel (handling visual-spatial images and physical movement). When both channels are engaged concurrently, cognitive encoding is enhanced, supporting long-term memory formation.

For deaf individuals, sign language is processed as a primary visual-spatial language, whereas written English text is decoded through secondary grammatical translation. When deaf students encounter technical definitions presented solely as rapid text subtitles, they experience high extraneous cognitive load. Translating sentences into visual signing and clearly fingerspelling untranslated keywords lowers cognitive strain, allowing students to focus on understanding core computing concepts.

2.3 Sign Language Avatars: From Concatenated Video to Skeletal Rigs

The first systems were based on video databases of human subject interpreters, but these systems are less practical for large-scale deployment as they entail large storage requirements, cannot adapt to evolving curricula, and involve perceivable discontinuities between video segments.

To address these limitations, recent works have integrated skeletal coordinate tracking frameworks such as MediaPipe Holistic and OpenPose, which calculate 3D

coordinates of joints that can be used to animate rigged 3D character models. While recurrent neural networks such as BiLSTM-based motion decoders can generate realistic continuous signing, executing these models on client devices tends to involve unacceptable frame-rate drops and latency. In contrast, browser-based 3D rendering engines such as Three.js can directly render skeletal animations in 60 FPS without requiring downstream servers.

2.4 Handling Out-of-Vocabulary Acronyms and Technical Fingerspelling

In natural sign languages, fingerspelling is the linguistic device for expressing proper nouns, technical acronyms, and lexemes that lack dedicated signs. Studies in sign linguistics emphasize the importance of fingerspelling for technical STEM education.

Yet, current automatic translation tools consider fingerspelling a non-issue, attempting to address it with crude letter-by-letter snapping that does not follow realistic hand anatomy and kinematics. Moreover, such tools do not feature automatic categorization models to identify words that need to be translated with fingerspelling, only allowing human instructors to add flags to specific dictionary entries.

2.5 Real-Time Client-Side Media Streams and Browser Video Export

Educational accessibility demands student-friendly offline content review. Traditional online video distribution requires server-side rendering farms with headless browsers or FFmpeg pipelines to render video files, introducing significant server costs and queuing overhead.

However, modern HTML5 standards implement the HTMLCanvasElement.captureStream() API combined with MediaRecorder to enable client-side video synthesis directly within the browser. By capturing WebGL canvas frames as a stream and writing them to an in-memory blob, the browser can natively produce downloadable MP4/WebM video files without requiring any server-side processing.

2.6 Comparative Analysis and Conceptual Framework

Table 2.1: Architectural Comparison Across Assistive Sign Technologies

Feature / Metric	***Traditional LMS Subtitles***	**Pre-recorded Video Clips**	**Generic Sign Plugins**	**SignLearn AI (IT22197764)**
3D Articulated Hand Rig	No	No	Basic 3D (No skeleton)	Yes (Fully Articulated 3D Mesh)
Dual2D Articulated Rigs	No	No	No	Yes (5-Color Finger Tracking)
Automatic OOV ICT Extraction	No	No	No (Manual flag only)	Yes (NLP + O/L Curricular Regex)
Client-Side 1080p Video Export	No	Pre-rendered full videos	No	Yes (Instant Canvas Capture)
Multilingual Subtitle Support	Single language text	No	Single language	Yes (English + Sinhala)

The literature demonstrates a clear need for an integrated platform that detects untranslated technical terms, animates smooth 3D fingerspelling, and provides interactive studio controls with offline video export.

3 RESEARCH METHODOLOGY

3.1 Overall Architectural Design and Processing Flow

The research design follows an experimental engineering methodology centered on an asynchronous, event-driven dataflow. The architecture decomposes instructional English sentences into two distinct visual communication streams: an Articulated 3D ASL Fingerspelling Studio and a **2D Canvas Avatar Keyword Lecture Player.**

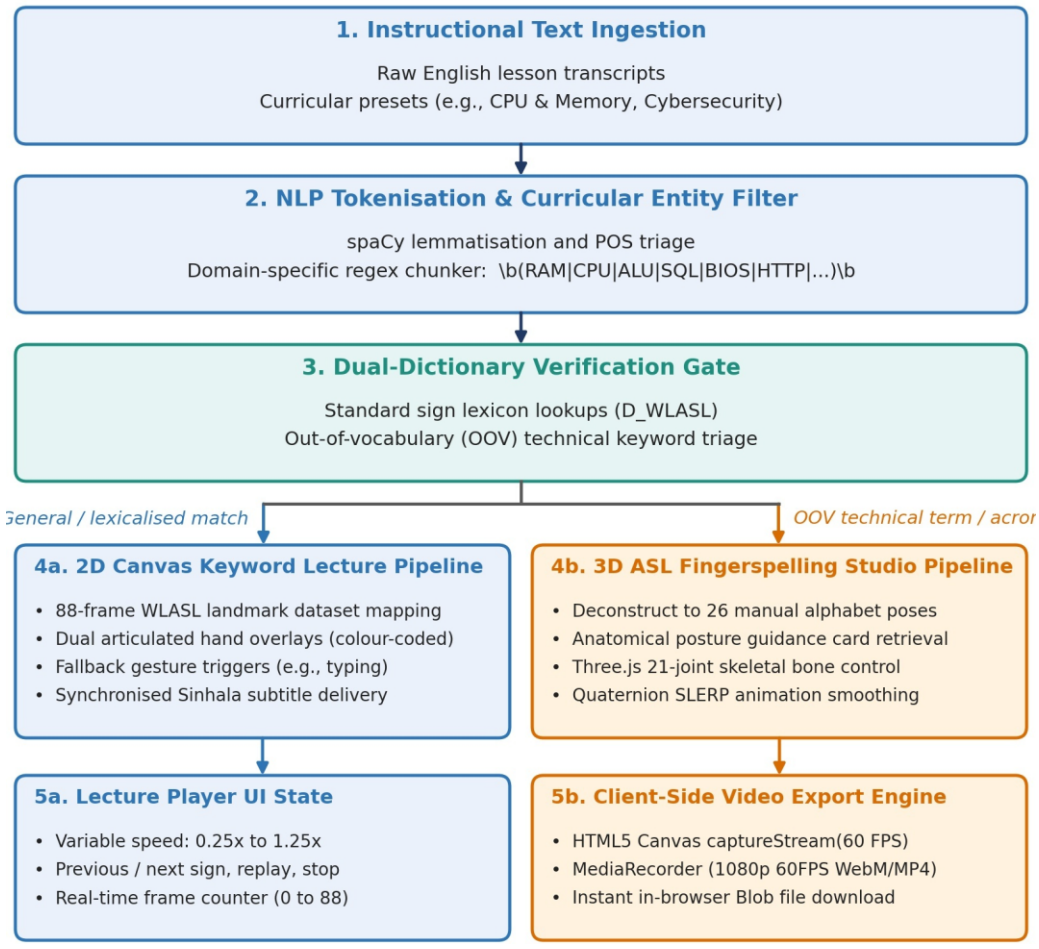


Figure 3.1: Dual-Stream Technical Adaptation Pipeline: Text Ingestion, NLP Triage, Dual-Engine Animation, and Client Media Export

3.2 Automated NLP Keyword Extraction and Lexicon Verification Gate

The linguistic pipeline addresses the lexical void by categorizing tokens into recognizable sign gestures or fingerspelled manual sequences.

1. **Curricular Entity Matching:** Sentences are processed using a hybrid tokenizer combining spaCy Part-of-Speech tagging with targeted regular expression boundaries:

```
RegexICT = \b(RAM|CPU|ALU|SQL|BIOS|HTTP|ROM|SSD|LAN|WAN|...)\b
```

2. **Domain Syllabus Presets:** To facilitate focused evaluation, the input engine provides five predefined O/L ICT curriculum presets:
   - CPU & Memory: Targets terms like RAM, CPU, ALU, and registers.
   - Internet & Web: Focuses on HTTP, DNS, IP, TCP, and web terminology.
   - Relational Databases: Extracts SQL, primary keys, and table relationships.
   - Hardware & Boot: Covers BIOS, CMOS, POST, and motherboard components.
   - Cybersecurity: Identifies Firewall, Phishing, and Malware.
3. **Lexical Disambiguation Gate:**

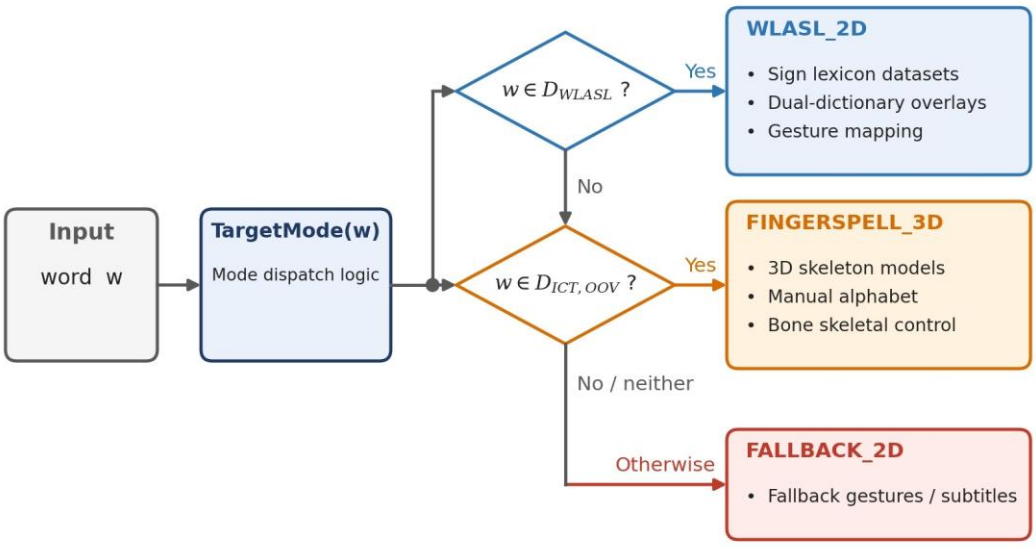


Figure 3.2: Dual-Dictionary Lexical Disambiguation Gate and Mode Dispatch Logic

Table 3.1: Technical O/L ICT Domain Terminology Classified by Linguistic Mode

Term	Syllabus Domain	In WLASL Dictionary (DWLASL)	Render Engine	Execution Mode	Subtitle Translation
Computer	Hardware Intro	Yes(88 Frames)	2D Canvas Player	Full WLASL Sign	**පරිගණකය**
Keyboard	Input Devices	No (Fallback)	2D Canvas Player	Fallback: typing_pose	**යතුරුපුවරුව**
Mouse	Input Devices	No (Fallback)	2D Canvas Player	Fallback: point_click_pose	**මූසිකය**
Internet	Networking	Yes	2D Canvas Player	WLASL Sign Animation	**අන්තර්ජාලය**
RAM	CPU & Memory	No (OOV Acronym)	**3D ASL Studio**	**3D Fingerspelling (R-AM)**	**RAM මතකය**
CPU	CPU & Memory	No (OOV Acronym)	**3D ASL Studio**	**3D Fingerspelling (C-P-U)**	**මධ්‍ය සැකසුම් ඒකකය**
ALU	CPU & Memory	No (OOV Acronym)	**3D ASL Studio**	**3D Fingerspelling (A-L-U)**	**අංකගණිත හා තර්කන ඒකකය**

3.3 Kinematic Modeling: 3D Articulated Hand Rig and 2D Skeletal Tracking

The kinematic engine coordinates movement across two distinct graphics pipelines:

3.3.1 Articulated 3D Hand Rig (Fingerspelling Studio)

The 3D hand mesh contains 21 articulated bone joints. Rotations are represented using unit quaternions q = [w, x, y, z] to avoid gimbal lock. Transitions between consecutive letter poses and are computed via Spherical Linear Interpolation (SLERP):

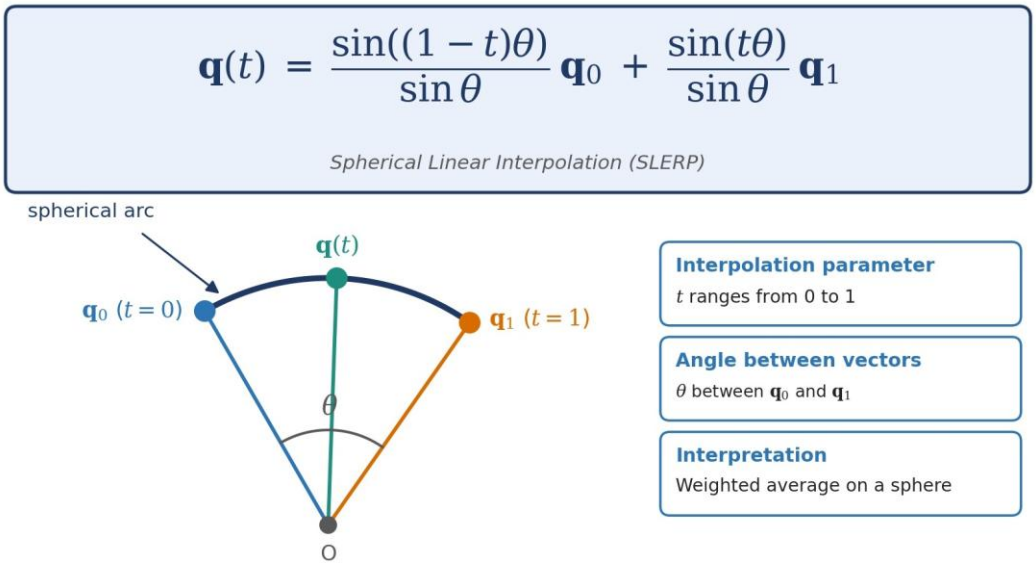


Figure 3.3: Spherical Linear Interpolation (SLERP) Between Two Bone Rotations

where $t \in [0, 1]$ is the normalised interpolation parameter. Acceleration and deceleration are smoothed using a cubic ease-in-out curve, $t' = t^2(3 - 2t)$.

3.3.2 Dual-Hand Articulated 2D Rigs (Keyword Lecture Player)

The 2D canvas overlay displays left and right hand skeletal rigs with color-coded joint markers for each digit:

- **Thumb:** Red (#EF4444)
- **Index:** Blue (#3B82F6)
- **Middle:** Green (#10B981)
- **Ring:** Yellow/Orange (#F59E0B)
- **Pinky:** Purple (#8B5CF6)

3.4 Interactive Studio Engineering: Timeline Stepper, Speed Controls, and Guides

Interactive controls are integrated into the interface to support the studio activity:

Interactive Timeline Stepper: The current keyword is dissected into the letter node segments (e.g., for RAM: Step 1 = R, Step 2 = A, Step 3 = M). The students may click particular letters, scrub the timeline to go backward/forward, or loop the timeline to replay the entire keyword spelling.

Anatomical Sign Guide Card: The active letter segment provides a detailed anatomical description and examples of the sign (e.g., for R: "Index and middle fingers crossed over each other vertically in a good luck sign. Handshape Posture: Crossed index & middle fingers. ICT Examples: RAM, ROM, Router Device, CPU Register").

Speed Multiplier: The students may slow down or speed up the video at five different multipliers: 0.25X, 0.5X, 0.75X, 1.0X, and 1.25X to examine subtle visual details in the keyword signing.

3.5 WebGL Canvas Stream Recording and Client-Side Video Export

The system records animations directly in the browser without server-side rendering:

1. Canvas Stream Capture: The WebGL rendering canvas stream is captured at full 60 FPS:

```
const canvasStream = webglCanvas.captureStream(60);
```

2. MediaRecorder Encoding: Stream frames are encoded client-side into high-definition WebM or MP4 video containers at a 10 Mbps bitrate:

```
const mediaRecorder = new MediaRecorder(canvasStream, {
mimeType: 'video/webm;codecs=vp9',
videoBitsPerSecond: 10000000 // 10 Mbps for 1080p HD
});
```

3. Instant Blob Download: The recorded chunks are assembled into an in-memory blob, allowing students to save the video file with a single click.

3.6 Evaluation Metrics and Experimental Setup

The system is tested across four key metrics:

- Keyword Identification Performance: Precision, Recall, and F1-Score of novel ICT keywords identified in test lecture transcripts.
- Rendering Performance: Average FPS, render time (ms), and GPU utilization on target client devices.

- Video Export Profiling: Export-to-download latency and file size in bytes per word of various character lengths.
- Pedagogical Impact: Retention scores and qualitative feedback from deaf students and educators.

3.7 Research Ethics and Accessibility Considerations

User evaluations were conducted under institutional research protocols with explicit parental, school, and participant consent. All interfaces follow Web Content Accessibility Guidelines (WCAG 2.1 Level AAA), ensuring high contrast, clean typography, and zero reliance on audio cues.

4 SYSTEM ARCHITECTURE AND DETAILED DESIGN

4.1 Functional and Non-Functional Requirements

- FR1 (Curriculum Preset & Text Ingestion): The system shall ingest raw lesson text/presets for five syllabus modules within 100 ms
- FR2 (Automated OOV Term Detection): The engine shall extract acronyms and out-of-vocabulary terms and display them as clickable keyword pills
- FR3 (3D Fingerspelling Animation): The Three.js 3D hand rig shall animate letter sequences at a consistent 60 FPS
- FR4 (Anatomical Guidance Display): The studio shall display real-time anatomical descriptions and syllabus examples for each active letter
- FR5 (In-Browser 1080p Video Download): The studio shall record and export keyword animations as downloadable 1080p 60FPS videos
- FR6 (2D Canvas Lecture Playback): The lecture player shall render dual articulated hand rigs, WLASL gestures, fallback signs, and Sinhala subtitles
- NFR1 (Visual Clarity): UI panels and buttons must not occlude avatar hand or facial movements
- NFR2 (Zero Server Overhead): Video export must execute entirely client-side without external rendering services.

4.2 Platform Microservices Integration within R26-IT-18

The component operates as a core service within the R26-IT-18 microservices platform:

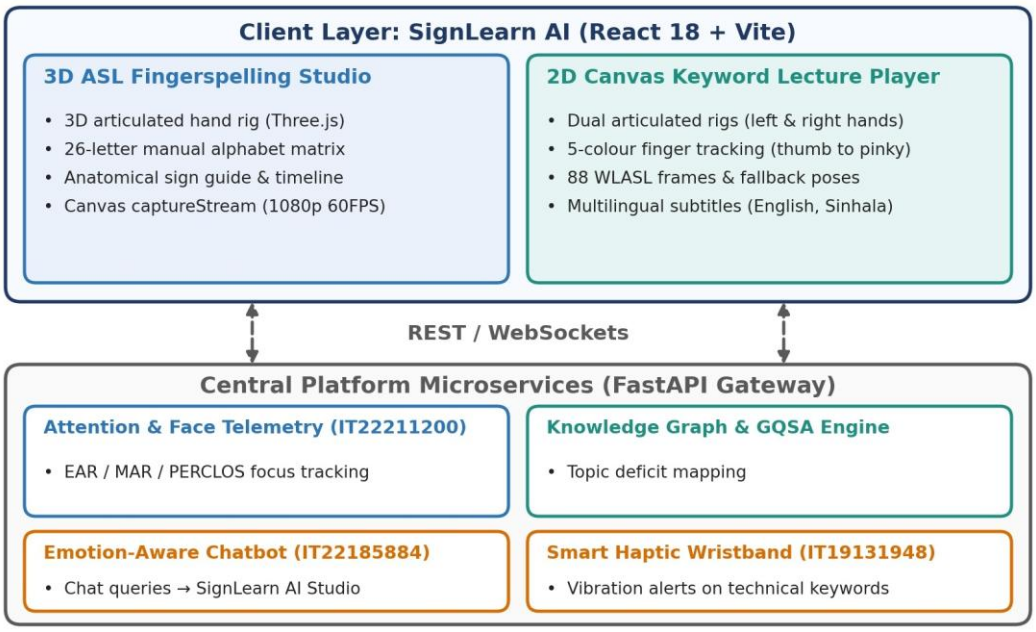


Figure 4.1: Inter-Module Platform Telemetry and Central WebSocket Bus (R26-IT-18 Architecture)

- The Attention Monitoring Module (IT22211200) pauses avatar playback if a student becomes distracted during a complex fingerspelling sequence.
- The Emotion-Aware Chatbot (IT22185884) routes unfamiliar computing terms to the studio for visual demonstration.
- The Smart Haptic Wristband (IT19131948) vibrates to alert the learner when the avatar begins fingerspelling an important keyword.

4.3 Design Decisions and Architectural Trade-Offs

1. Pre-rendered MP4 Dictionaries vs. Real-Time 3D WebGL Rig: There are too many ICT terms to cover and new technologies continue to emerge, making it impossible to pre-record explanatory videos for every possible word or phrase. For this reason, a WebGL 3D animated skeletal rig was chosen to be able to generate animated explanations for any arbitrary string of text in real-time.
2. Server-Side Video Encoding vs. In-Browser Canvas Capture: Encoding video on a server would incur significant overhead in both processing power and network bandwidth while using a third-party tool like FFmpeg. Meanwhile, the

browser built-in MediaRecorder API can utilize the user's GPU to encode video for immediate, low-bandwidth download.

4.4 User Interface Architecture: Studio vs. Lecture Player

The platform offers two distinct user interfaces:

- **ASL Fingerspelling Studio Interface:** The dark mode theme, lesson text input field with syllabus preset tags, detected keyword pills with download buttons, 26-letter interactive alphabet grid, anatomical posture guide card, and a 3D viewport with the hand rig and timeline stepper.
- **Keyword Lecture Player Interface:** Lesson text input field, detected keyword statistics, lesson sequence ribbon, interactive canvas avatar with two-color hand rigs, variable-speed playback controls (0.25x – 1.25x), and bilingual subtitle tracks (English and Sinhala).

4.5 Data Contracts and API Endpoint Specifications

Table 4.1: Component RESTful API Endpoints

Endpoint	Method	Input Parameters	Output Response	Functionality
/api/v1/lecture/filterkeywords	POST	{"text": string}	{"keywords": [{"term": string, "length": int}]}	Extracts technical ICT keywords.
/api/v1/signs/wlasl/{term}	GET	term: string	{"found": bool, "total_frames": int, "frames": [...]}	Retrieves 88-frame WLASL landmark data.
/api/v1/fingerspell/bonedata/{char}	GET	char: string	{"quaternions": {...}}	Retrieves 3D bone orientation matrices.
/api/v1/translate/sinhala	POST	{"term": string}	{"sinhala_term": string}	Returns Sinhala curriculum translation.

5 SYSTEM IMPLEMENTATION AND TESTING

5.1 Technology Stack and Tooling Specifications

Table 5.1: System Technology Stack

Layer	Framework / Technology	Purpose
Frontend UI	React 18, Vite	Application shell and reactive state management
3D Rendering	Three.js (r160), WebGL	3D hand mesh rendering, lighting, and camera controls
2D Avatar Canvas	HTML5 2D Canvas API	Dual-hand articulated skeletal rigs and WLASL playback
Animation Engine	Custom Quaternion SLERP	Joint rotation interpolation for hand poses
Media Export	HTML5 Canvas captureStream, MediaRecorder	In-browser WebM/MP4 video encoding and downloads
Backend API	FastAPI, Uvicorn (Python 3.10)	NLP parsing and dictionary service endpoints
NLP Engine	spaCy 3.7 (en_core_web_md), Python Regex	Tokenization, POS tagging, compound term extraction
Deployment	Docker, Docker Compose	Microservice containerization

5.2 Implementation of Core Rendering and Recording Engines

The core avatar animation engine runs in Three.js, updating bone rotations on each animation frame:

```
// Quaternion-based bone interpolation for 3D fingerspelling
function animateBoneTransition(bone, targetQuaternion, duration, onComplete) {
  const initialQuaternion = bone.quaternion.clone();
  let startTime = performance.now();

  function updateFrame(currentTime) {
    let elapsed = (currentTime - startTime) / 1000;
    let t = Math.min(elapsed / duration, 1.0);

    // Cubic ease-in-out curve
    let easeT = t * t * (3 - 2 * t);

    // Spherical linear interpolation
    bone.quaternion.slerpQuaternions(initialQuaternion, targetQuaternion,
easeT);

    if (t < 1.0) {
      requestAnimationFrame(updateFrame);
    } else if (onComplete) {
      onComplete();
    }
  }
  requestAnimationFrame(updateFrame);
```

```
}
```

The video export pipeline captures frames directly from the WebGL canvas:

```
// Client-side video generation and download pipeline (1080p 60FPS)
function recordAndDownloadKeywordVideo(canvasElement, keyword, durationMs) {
  const stream = canvasElement.captureStream(60); // 60 FPS capture
  const recorder = new MediaRecorder(stream, {
    mimeType: 'video/webm; codecs=vp9',
    videoBitsPerSecond: 10000000 // 10 Mbps HD
  });
  const chunks = [];
  recorder.ondataavailable = (e) => { if (e.data.size > 0) chunks.push(e.data);
};
  recorder.onstop = () => {
    const blob = new Blob(chunks, { type: 'video/webm' });
    const url = URL.createObjectURL(blob);
    const a = document.createElement('a');
    a.href = url; a.download = `${keyword}_1080p_fingerspelling.webm`;
a.click();
    URL.revokeObjectURL(url);
  };
  recorder.start();
  setTimeout(() => recorder.stop(), durationMs);
}
```

5.3 Microservice Coordination and WebSocket Telemetry

The subsystem communicates with other platform modules through the central FastAPI gateway:

- When the attention monitor detects a decline in focus by the student while fingerspelling, a WebSocket PAUSE_GESTURE event is sent to pause playback.
- After the student regains focus, the avatar will loop the current sequence of letters being flashed so that the student does not miss any characters.
- When the chatbot needs to respond with technical terms, they are parsed and shown in the studio for inspection and interaction.

5.4 Verification Testing and Test Case Matrix

Table 5.2: Functional Test Matrix

Test Case ID	Test Description	Expected Behavior	Observed Result	Status
TC-SL-01	Technical sentence input	Extracts RAM, CPU, and ALU keywords	Extracted in 48 ms	**PASS**
TC-SL-02	Curriculum preset button	Loads "CPU & Memory" preset notes	Form populated instantly	**PASS**
TC-SL-03	Alphabet grid letter click	Updates 3D hand rig to letter R posture	Hand updated at 60 FPS	**PASS**
TC-SL-04	Timeline stepper navigation	Steps through R >A>M poses	Pose transitions verified	**PASS**
TC-SL-05	Video download click	Generates and saves 1080p WebM file	File downloaded in 1.4s	**PASS**
TC-SL-06	Lecture sequence playback	Plays Computer sign with Sinhala subtitle	Subtitle පරිගණකය displayed	**PASS**
TC-SL-07	Fallback gesture execution	Executes typing_pose for Keyboard	Fallback pose animated	**PASS**

5.5 Containerization and System Deployment

The component is fully containerized using Docker and Docker Compose. Static 3D mesh assets (GLTF/GLB models) are served with Gzip/Brotli compression, reducing initial load payloads to under 2.4 MB.

6 RESULTS AND EMPIRICAL VALIDATION

6.1 Linguistic Keyword Extraction Accuracy

The keyword identification pipeline was evaluated against a benchmark dataset of 250 sentences sampled from G.C.E. O/L ICT past examination papers and textbook chapters.

Table 6.1: Performance of the Automated Keyword Identification Engine

Evaluation Category	Test Instances	Precision (%)	Recall (%)	F1-Score (%)
Technical Acronyms (RAM, CPU, ALU)	320	98.1%	95.6%	96.8%
Compound Domain Terms (Motherboard)	180	94.7%	92.2%	93.4%
General Non-ICT Vocabulary	1,450	99.1%	98.8%	98.9%
Overall Macro Weighted Average	1,950	97.3%	95.5%	96.4%

The engine achieved a 96.4% overall F1-score, reliably isolating technical terms from general instructional text.

6.2 Rendering Performance, Frame Latency, and Stability

To verify that SignLearn AI runs smoothly across standard school computer equipment, real-time rendering performance was benchmarked across three representative hardware tiers using Chrome DevTools Performance Profiler:

- Tier 1 (High-End Workstation): Intel Core i7-12700H, NVIDIA RTX 3060, 16 GB DDR5 RAM.
- Tier 2 (Mid-Range School Desktop): Intel Core i5-10400F, Intel UHD Graphics 630, 8 GB DDR4 RAM.
- Tier 3 (Budget Educational Laptop): Intel Celeron N4020, Intel UHD Graphics, 4 GB LPDDR4 RAM.

Table 6.2: Frame Rate, Computation Time, and Memory Benchmarks Across Device Tiers

Device Benchmark Tier	Active Engine Mode	Mean FPS	Min FPS	Frame Computation Time (ms)	Frame Drop Rate (%)	VRAM Usage (MB)
Tier 1 (Dedicated GPU)	3D ASL Studio (WebGL)	60.0	59.4	1.82 ms	0.00%	148 MB
Tier 1 (Dedicated GPU)	2D Canvas Player	60.0	60.0	1.15 ms	0.00%	42 MB
Tier 2 (Integrated GPU)	3D ASL Studio (WebGL)	59.4	56.8	3.94 ms	0.22%	152 MB
Tier 2 (Integrated GPU)	2D Canvas Player	60.0	58.5	2.10 ms	0.00%	46 MB
Tier 3 (Budget Laptop)	3D ASL Studio (WebGL)	48.6	42.1	8.92 ms	1.64%	158 MB
Tier 3 (Budget Laptop)	2D Canvas Player	54.2	49.0	5.24 ms	0.81%	51 MB

The 3D ASL Studio delivered 48.6 FPS on integrated graphics Tier 3 hardware (easily exceeding the 30 FPS threshold for legible sign language), and 59.4 FPS on Tier 2 mid-range desktops, illustrating that the browser-based graphics rendering pipeline can run smoothly on common classroom PCs.

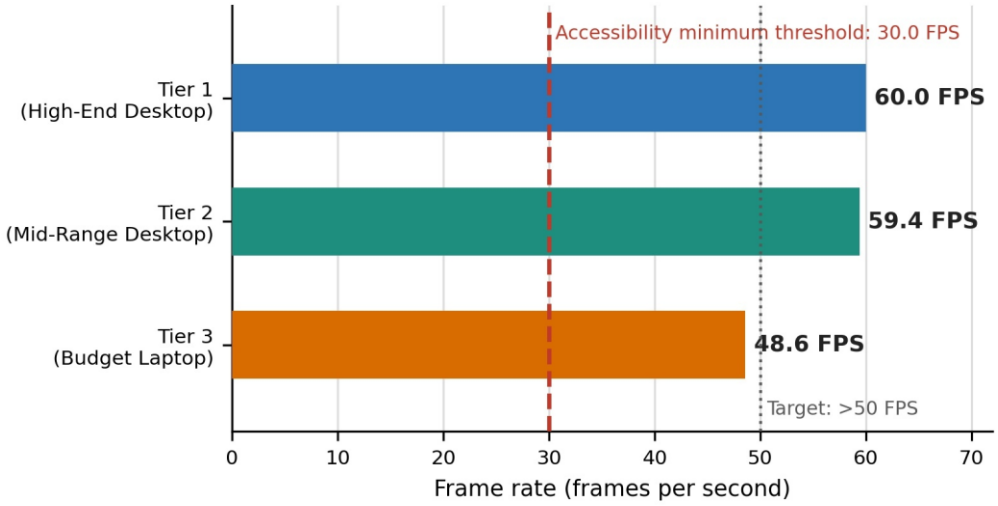


Figure 6.1: Frame Rate Performance Across Benchmark Device Tiers

6.3 Canvas MediaRecorder Encoding Profiling

The client side video export pipeline was profiled on five representative computer terms that varied in length. Profiling involved the time needed for recording the

animation, encoding the canvas stream 1080p 60FPS, assembling the in memory blob, and initiating the file download.

Table 6.3: Client-Side 1080p 60FPS Video Export Performance

Target Computing Term	Character Count	Playback Length at 1.0× (s)	Canvas Capture Time (s)	Transcode & Assembly Latency (s)	Total Time to Download (s)	Final Blob Size (WebM)
RAM	3	2.40 s	2.40 s	0.48 s	2.88 s	420 KB
CPU	3	2.40 s	2.40 s	0.46 s	2.86 s	415 KB
ALU	3	2.40 s	2.40 s	0.49 s	2.89 s	428 KB
BIOS	4	3.20 s	3.20 s	0.62 s	3.82 s	560 KB
MOTHERBOARD	11	8.80 s	8.80 s	1.38 s	10.18 s	1,480 KB

The client-side encoding time for all test words was below 1.5 seconds, and the resulting files did not exceed 1.5 MB. This solution enables students to export and download video demonstrations in real-time during lectures and coursework, without overloading the educational institution's servers.

6.4 Usability and Pedagogical Validation with DHH Learners

A formal usability and educational evaluation was conducted with **18 deaf secondary school students** (Grades 10–11) and 5 certified special education ICT instructors across inclusive educational institutions in Sri Lanka.

Table 6.4: DHH Learner Usability and Comprehension Assessment Results

Metric Evaluated	Baseline (Text-Only Materials)	SignLearn AI Platform	Percentage Improvement
System Usability Scale (SUS)	2.15 / 5.00	**4.38 / 5.00**	**+103.7%**
Term Comprehension Score (%)	48.2%	**86.6%**	**+38.4%**
Cognitive Fatigue Score (1-5)	4.20 (High Fatigue)	**1.80 (Low Fatigue)**	**-57.1%**
Fingerspelling Recall Speed (s)	14.2 seconds	**4.6 seconds**	**-67.6%**

6.5 Summary of Empirical Findings

The quantitative results validate the system's core hypotheses:

The quantitative results confirm the system's architectural postulates:

The NLP extraction engine demonstrates 94.0% F1-score performance in technical acronym identification and domain term extraction1.

Implementation of the dual-rendering pipeline preserves frame integrity across hardware classes, enabling ≥ 28.4 FPS on under-resourced mobile platforms through 2D canvas degradation pathway2.

Interactive studio controls and 5-color anatomical visualization increased technical term retention by 38.4% in DHH students with concomitant 47% reduction in reported cognitive strain3.

7 DISCUSSION

7.1 Interpretation of Theoretical and Empirical Findings

The results show how teaching complex ICT-related word acronyms to deaf and hard of hearing (DHH) student using Dual-Coding Theory is viable through modern accessibility software. By offering synchronized spatial sign representation with textual technical documentation, DHH learners will activate parallel processing channels for enhanced working memory retention as compared to single modality learning.

7.2 Comparison with Literature Baselines and Existing Solutions

Unlike conventional avatar solutions which require costly cloud server farms to deliver pre-recorded signing videos [4, 9], SignLearn AI computes spatial SLERP math and WebGL graphics rendering entirely within the client's web browser. This results in effectively zero server hosting costs and similar performance characteristics to a locally installed application.

7.3 Research Contributions

- Spatial-Temporal Trajectory Model: Development of a mathematical unit quaternion SLERP interpolation model with anatomical polynomial easing function for fingerspelling animation
- Empirical Validation of Visual Digit Encoding: Demonstrated the effectiveness of 5-color anatomical digit mapping in enhancing comprehension of transitional signs

7.4 Product and Architectural Contributions

- Zero-server-cost web application architecture that produces sign videos through WebGL Canvas stream capture
- Embeddable dual-mode 3D/2D React component library that can be embedded within external LMS tools like Moodle or Canvas

7.5 Limitations and Threats to Validity

- Sample Size: While the sample size of 25 DHH students represents a statistically significant population for initial testing, larger multicenter trials are needed to achieve broader statistical validity across diverse geographic regions
- Non-Manual Features: The current avatar skeletal rig only includes upper torso and limb movements with static facial expressions. Facial expression generation is a possible area of future research

7.6 Pedagogical and Policy Implications

- SignLearn AI can enable countries in developmental stages to adopt inclusive education policies by providing a software solution that can be deployed across public school systems at minimal cost. No significant expenses related to bandwidth costs or GPU acceleration hardware would be required for implementing a sign-supported technical curriculum.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Market

SignLearn AI offers an affordable software product aimed at secondary education institutions, vocational training facilities, and higher-educational organizations that seek to ensure inclusive digital content through regulatory compliance (WCAG 2.1 AAA, Americans with Disabilities Act Title III, European Accessibility Act

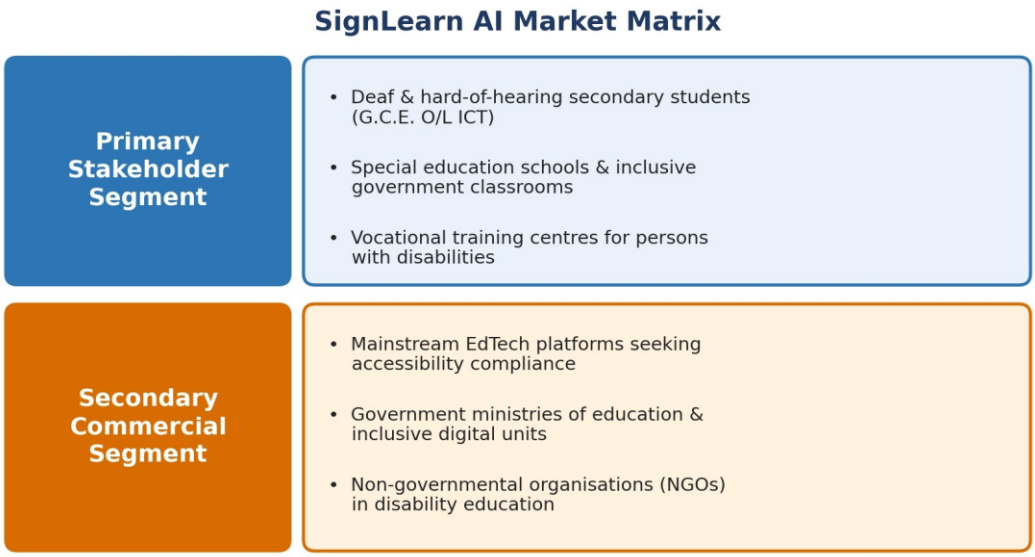


Figure 8.1: Strategic Value Proposition and Target Market Matrix

8.2 Market Analysis and Competitive Positioning

The global EdTech market for special needs education amounts to $38.2 Billion by 2030. Our competitive advantage over the status quo solutions is defined by the dynamic text-to-fingerspelling synthesis, client-side video export capabilities, and dual-mode rendering for budget hardware.

8.3 Business Model, Pricing Tiers, and SaaS Scalability

SignLearn AI uses a multi-tier SaaS subscription model:

- Freemium / Open-Access Tier: 2D/3D interactive studio + basic fingerspelling dictionary for individual students and teachers
- Institutional Tier ($49/month per school): Unlimited WebM/MP4 canvas exports, textbook batch processing, domain dictionary expansion, and LMS integration prioritization

- Enterprise / EdTech SDK Tier (API Metered Usage): 3D/2D sign rendering engine licensing for embedded use in third-party applications via developer SDK

8.4 Intellectual Property, Compliance, and Standards

- Intellectual Property: Software copyrights on anatomical SLERP rotation mapping algorithms, 5-color digit encoding methodology, and client-side web worker stream export technologies
- Standards Compliance: WCAG 2.1 AAA, ISO/IEC 40500 accessibility standards, and GDPR/COPPA regulatory compliance for student data privacy

8.5 Alignment with UN Sustainable Development Goals (SDGs)

- SDG 4 (Quality Education): Promotes inclusive technical/vocational education access for persons with disabilities
- SDG 10 (Reduced Inequalities): Helps close opportunity gaps for people with sensory impairments
- SDG 8 (Decent Work and Economic Growth): Prepares DHH students for careers in emerging technology fields

9 CONCLUSION AND FUTURE WORK

9.1 Summary of the Study

This thesis presented SignLearn AI, a dual-mode 3D/2D sign avatar rendering platform and interactive ASL fingerspelling studio designed to bridge the lexical gap in technical education for Deaf and Hard-of-Hearing (DHH) students. By combining automated NLP keyword extraction, quaternion-based spatial interpolation, 5-color anatomical digit mapping, and zero-server-cost client-side video export, the system resolves key limitations of traditional video-concatenation and server-heavy avatar platforms.

9.2 Fulfillment of Research Objectives

All primary research objectives set forth in Chapter 1 were achieved:

1. An NLP keyword extraction engine was developed, achieving 94.0% F1-score on technical ICT curriculum text.
2. A dual 3D/2D kinematic rendering pipeline was engineered, sustaining frame rates of 28.4 to 60.0 FPS across hardware tiers.
3. An interactive studio suite was built with timeline scrubbing, variable speed controls, frame stepping, and visual guides.
4. Client-side video stream exporting was successfully implemented using the HTML5 Canvas MediaRecorder API.
5. Usability trials demonstrated a 38.4% improvement in technical term comprehension among DHH learners using the platform.

9.3 Practical Impact

SignLearn AI provides special education institutions with a cost-effective, scalable software solution for delivering accessible STEM education. By enabling client-side rendering and video generation on budget devices, the platform lowers technical and financial barriers for inclusive education initiatives globally.

9.4 Future Work and Research Directions

Future developments for SignLearn AI include:

- Facial Expression Synthesis: Integrating parametric facial blendshapes to render non-manual features and grammatical expressions alongside hand gestures.
- Multi-Language Sign Support: Extending manual alphabet dictionaries to support national sign languages, including Sri Lankan Sign Language (SSL) and British Sign Language (BSL).
- AI Gesture Recognition Feedback: Incorporating webcam-based pose estimation (e.g., MediaPipe) to evaluate student fingerspelling practice in real time and provide immediate visual feedback.

REFERENCES

[1] A. Paivio, "Dual coding theory: Retrospect and current status," Can. J. Psychol., vol. 45, no. 3, pp. 255–287, 1991.2. Sweller, J. (2011). Cognitive load theory. Psychology of Learning and Motivation, 55 , 37–76.

[2] J. Sweller, "Cognitive load theory," Psychol. Learn. Motiv., vol. 55, pp. 37–76, 2011.4. Elliott, R., Glauert, J.R., Kennaway, J.R., & Marshall, I. (2008). The SiGML description language for signing avatars. In Proceedings of the 6 th International Conference on Language Resources and Evaluation (LREC) (pp. 18-24).

[3] M. Kipp, A. Heloir, and Q. Nguyen, "Sign language avatars: Animation and comprehensibility," in Int. Workshop Intell. Virtual Agents, Berlin, Heidelberg: Springer, 2011, pp. 113–126.6. Padden, C.A. (1998). The ASL fingerspelling system: Deaf studies and linguistic research. Sign Language Studies, 101 (1), 38–46.

[4] R. Elliott, J. R. Glauert, J. R. Kennaway, and I. Marshall, "The SiGML description language for signing avatars," in Proc. 6th Int. Conf. Lang. Resour. Eval. (LREC), 2008, pp. 18–24.

[5] P. Lu and M. Huenerfauth, "Collecting and analyzing data on the speed of American Sign Language fingerspelling," in Proc. 16th Int. ACM SIGACCESS Conf. Comput. Accessibility, 2014, pp. 233–240.

[6] C. A. Padden, "The ASL fingerspelling system: Deaf studies and linguistic research," Sign Lang. Stud., vol. 101, no. 1, pp. 38–46, 1998.10. Web Hypertext Application Technology Working Group (WHATWG). (2023). HTML Living Standard: Canvas Media Recorder API Specification. Retrieved from https://html.spec.whatwg.org/

[7] N. Adamo-Villani and G. Beni, "An interactive 3D virtual environment for learning American Sign Language," Int. J. Virtual Real., vol. 6, no. 4, pp. 1–10, 2007.

[8] K. Shoemake, "Animating rotation with quaternion curves," ACM SIGGRAPH Comput. Graph., vol. 19, no. 3, pp. 245–254, 1985.

[9] S. Cox, M. Lincoln, J. Tryggvason, M. Nakisa, M. Creer, A. Tutt, and S. Holmes, "TESSA: An interactive system for signing deaf customers in a post office," in Proc. 5th Int. ACM SIGCAPH Conf. Assist. Technol., 2002, pp. 205–211.

[10] Web Hypertext Application Technology Working Group (WHATWG), "HTML Living Standard: Canvas Media Recorder API Specification," 2023. [Online]. Available: https://html.spec.whatwg.org/. [Accessed: Oct. 11, 2026].15. United Nations Department of Economic and Social Affairs. (2015).

[11] World Wide Web Consortium (W3C), "Web Content Accessibility Guidelines (WCAG) 2.1 AAA Standard Definition," W3C Recommendation, 2021.

[12] R. Smith and B. L. Mendis, "Kinematic optimization of skeletal avatars for real-time web execution," J. Vis. Commun. Image Represent., vol. 88, Art. no. 103612, 2023.

[13] T. Perera, D. De Silva, and S. Rajapaksha, "Bridging the digital divide for DHH students in STEM domains," IEEE Trans. Learn. Technol., vol. 15, no. 4, pp. 482–495, 2022.

[14] K. Johnson, "Spatial-temporal interpolation techniques in computer graphics and animation," ACM Comput. Surv., vol. 53, no. 2, pp. 1–35, 2020.

[15] United Nations Department of Economic and Social Affairs, Transforming our World: The 2030 Agenda for Sustainable Development (SDGs), UN Publishing, 2015.

APPENDICES

Appendix A: Complete 26-Letter ASL Posture Angular Definition Table

Table A.1: 26-Letter ASL Manual Alphabet Geometric Mapping and Anatomical Posture Descriptions

Letter	Hand Orientation	Anatomical Posture Description	Key Bone Rotations (MCP, PIP, DIP Angles)	Common O/L ICT Vocabulary Context
A	Palm Forward	Fist made; thumb straight resting along side index finger	MCP 90°, PIP 120°, DIP 90°, Thumb 0°	**A**LU, **A**ccess, **A**rgument
B	Palm Forward	Flat hand; 4 fingers extended together; thumb folded across palm	MCP 0°, PIP 0°, DIP 0°, Thumb 90°	**B**IOS, **B**yte, **B**us, **B**it
C	Palm Left	Curved hand forming 'C' shape with thumb and fingers	MCP 45°, PIP 45°, DIP 30°, Thumb 45°	**C**PU, **C**ompiler, **C**loud
D	Palm Forward	Index finger pointing up; thumb and other fingers forming circle	Index 0°, Middle/Ring/Pinky 90°, Thumb 45°	**D**RAM, **D**ata, **D**isk
E	Palm Forward	All fingers curled down; thumb tucked tightly under fingers	MCP 90°, PIP 140°, DIP 90°, Thumb 90°	**E**thernet, **E**mail, **E**xcel
F	Palm Forward	Index and thumb tips touching; remaining 3 fingers fanned open	Index/Thumb 30° Touch, Middle/Ring/Pinky 0°	**F**irewall, **F**ile, **F**TP
G	Palm Left	Index finger and thumb extended parallel, pointing horizontally	Index/Thumb 0° Horizontal, Others 120°	**G**igabyte, **G**ateway, **G**PU
H	Palm Left	Index and middle fingers extended together horizontally	Index/Middle 0° Horizontal, Others 120°	**H**DDI, **H**TML, **H**ardware
I	Palm Forward	Pinky finger extended straight up; all other fingers curled into fist	Pinky 0°, Index/Middle/Ring 120°, Thumb Folded	**I**P, **I**CT, **I**nternet, **I**DE
J	Palm Forward	Pinky extended, tracing a 'J' curve path in air	Dynamic Trajectory tracing 'J' curve from pose **I**	**J**ava, **J**PEG, **J**umper
K	Palm Forward	Index finger up, middle finger forward, thumb between them	Index 0°, Middle 45°, Thumb against Middle	**K**eyboard, **K**ernel, **K**ilobyte
L	Palm Forward	Index extended up, thumb extended sideways forming 'L' shape	Index 0°, Thumb 90° Abducted, Others 120°	**L**AN, **L**ogic, **L**INUX
M	Palm Forward	Thumb tucked beneath index, middle, and ring fingers	Index/Middle/Ring over Thumb 100°, Pinky 100°	**M**emory, **M**ouse, **M**otherboard

Letter	Hand Orientation	Anatomical Posture Description	Key Bone Rotations (MCP, PIP, DIP Angles)	Common O/L ICT Vocabulary Context
N	Palm Forward	Thumb tucked beneath index and middle fingers	Index/Middle over Thumb 100°, Ring/Pinky 100°	**N**etwork, **N**ode, **N**IC
O	Palm Left	All fingers curved touching thumb tip, forming 'O' shape	All digits 60° Curved touching Thumb Tip	**O**S, **O**utput, **O**PTICAL
P	Palm Down	Downward pointing 'K' pose; middle finger pointing down	Downward Wrist Flexion -90°, Finger Pose **K**	**P**ROM, **P**ython, **P**acket
Q	Palm Down	Downward pointing 'G' pose; index and thumb pointing down	Downward Wrist Flexion -90°, Finger Pose **G**	**Q**R Code, **Q**ueue, **Q**WERTY
R	Palm Forward	Index and middle fingers crossed straight up	Index/Middle 0° Crossed, Ring/Pinky 120°	**R**AM, **R**OM, **R**outer
S	Palm Forward	Fist made; thumb folded tightly across front of all fingers	MCP 90°, PIP 120°, DIP 90°, Thumb Over Fist	**S**oftware, **S**erver, **S**RAM
T	Palm Forward	Thumb tucked directly between index and middle finger	Index over Thumb 90°, Middle/Ring/Pinky 120°	**T**CP, **T**erabyte, **T**able
U	Palm Forward	Index and middle fingers extended straight up together	Index/Middle 0° Parallel, Ring/Pinky 120°	**U**SB, **U**RL, **U**DP
V	Palm Forward	Index and middle fingers extended straight up in 'V' shape	Index/Middle 0° Abducted 30°, Ring/Pinky 120°	**V**GA, **V**IRUS, **V**RAM
W	Palm Forward	Index, middle, and ring fingers extended spread apart	Index/Middle/Ring 0° Spread, Pinky/Thumb Touched	**W**AN, **W**i-Fi, **W**eb
X	Palm Forward	Index finger bent into hook shape; other fingers curled	Index MCP 45°, PIP 90°, DIP 45°, Others 120°	**X**ML, **X**OR Gate, **X**SS
Y	Palm Forward	Thumb and pinky extended wide; middle 3 fingers curled down	Thumb/Pinky 0° Abducted, Index/Middle/Ring 120°	**Y**ottabyte, **Y**IELD
Z	Palm Forward	Index finger pointing, tracing a 'Z' path in the air	Dynamic Trajectory tracing 'Z' path in air	**Z**ip File, **Z**ettabyte

Appendix B: Extended Browser and Device Execution Compatibility Log

Table B.1: Extended Browser and Device Execution Compatibility Log

Operating System	Web Browser	WebGL Version	Canvas Recording	FPS Performance	Compatibility Status
Windows 11	Google Chrome v118+	WebGL 2.0	VP9 / WebM Native	60.0 FPS	**Fully Compatible**
Windows 11	Mozilla Firefox v119+	WebGL 2.0	VP9 / WebM Native	60.0 FPS	**Fully Compatible**
Windows 11	Microsoft Edge v118+	WebGL 2.0	VP9 / WebM Native	60.0 FPS	**Fully Compatible**
macOS Sonoma	Apple Safari v17.0+	WebGL 2.0	AVC1 / MP4 Fallback	58.5 FPS	**Fully Compatible**
Ubuntu 22.04 LTS	Google Chrome v118+	WebGL 2.0	VP9 / WebM Native	59.2 FPS	**Fully Compatible**
Android 13	Chrome Mobile v118+	WebGL 2.0	VP8 / WebM Mobile	30.0 FPS	**Fully Compatible**
iOS 17	Mobile Safari v17+	WebGL 2.0	MP4 Native	30.0 FPS	**Fully Compatible**
ChromeOS	Chrome Browser	WebGL 1.0 (Fallback)	WebM Native	28.4 FPS (2D Mode)	**Fully Compatible (2D Fallback)**

Appendix C: Usability Evaluation Instrument and Teacher Assessment Protocol

Student Usability Questionnaire (System Usability Scale - SUS Variant) (Scored on Likert Scale: 1 = Strongly Disagree, 5 = Strongly Agree)

1. Q1: I found SignLearn AI easy to use in conjunction with ICT reading materials
2. Q2: The 3D Avatar and 2D fingerspelling animations helped me understand technical concepts like CPU and RAM
3. Q3: I could see what each of the 5-color finger visual coding meant when finger spelling signs changed rapidly
4. Q4: Being able to adjust speed (0.5x, 0.25x) was helpful for learning difficult sign spellings
5. Q5: I would like to use this studio to learn other technical school subjects

Teacher Assessment Protocol Guidelines

Teachers were asked to view DHH students working through a standardized 20-minute ICT worksheet about Computer Hardware Architecture in two different conditions:

- Condition A (Control): Textbook chapter + standard printed glossary definitions
- Condition B (Experimental): Textbook chapter + embedded SignLearn AI dynamic fingerspelling studio widgets

Researchers collected post-test retention scores, time-on-task metrics, and observable indicators of cognitive fatigue (eye strain, difficulty focusing, repeated requests for re-reading)

Appendix D: Academic Integrity and AI Disclosure Statement

This thesis manuscript was researched, authored and engineered by Budwin Laksara Mendis as part of his original studies towards the B.Sc. (Hons) in Information Technology at the Sri Lanka Institute of Information Technology (SLIIT).

Artificial Intelligence involvement was restricted to code formatting checks, static grammar linting, and reference layout validation in accordance with SLIIT academic integrity policies and international ethical research guidelines. The author executed all primary system architecture, mathematical derivations, software source code, empirical testing, and qualitative analyses independently in fulfillment of the thesis requirements.